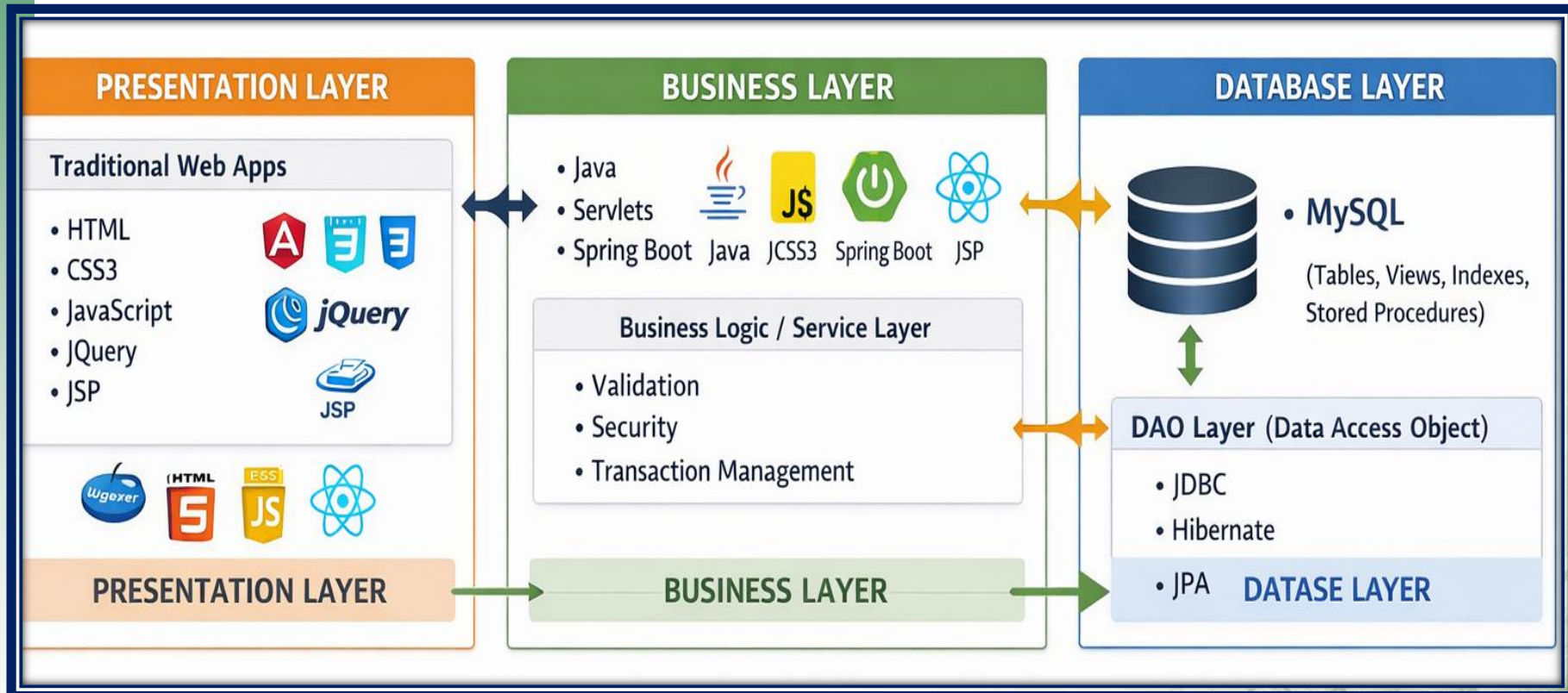




Session 1

Web Applications – An Overview

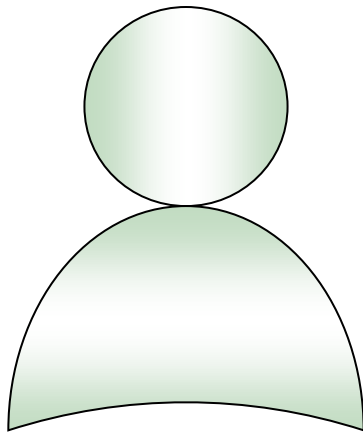
3-Tier Architecture Software Design Pattern



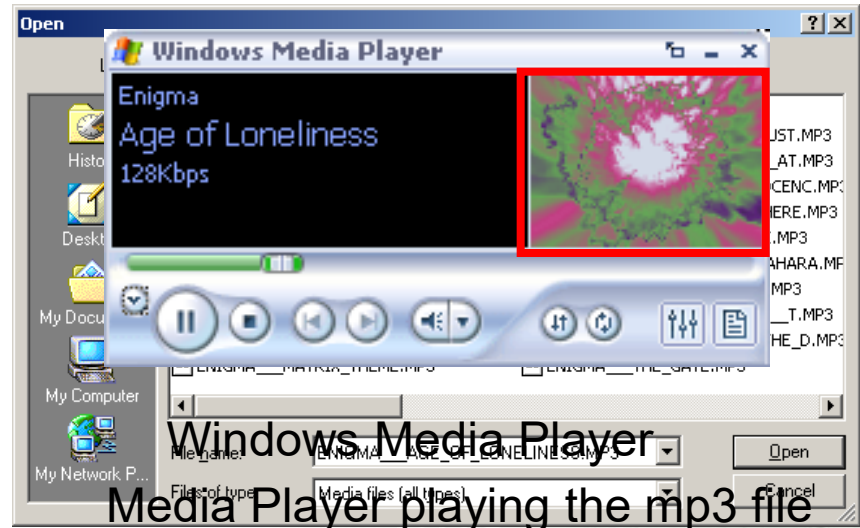
Objectives

- ❑ *Describe a multi-level architectural view of a Web application.*
- ❑ *Review HTTP protocol*
- ❑ *Describe the anatomy of a typical Web Application.*
- ❑ *Describe the Web development process*
- ❑ *Setup Tomcat environment in Eclipse/STS/VSCode*

Desktop Applications 2-1

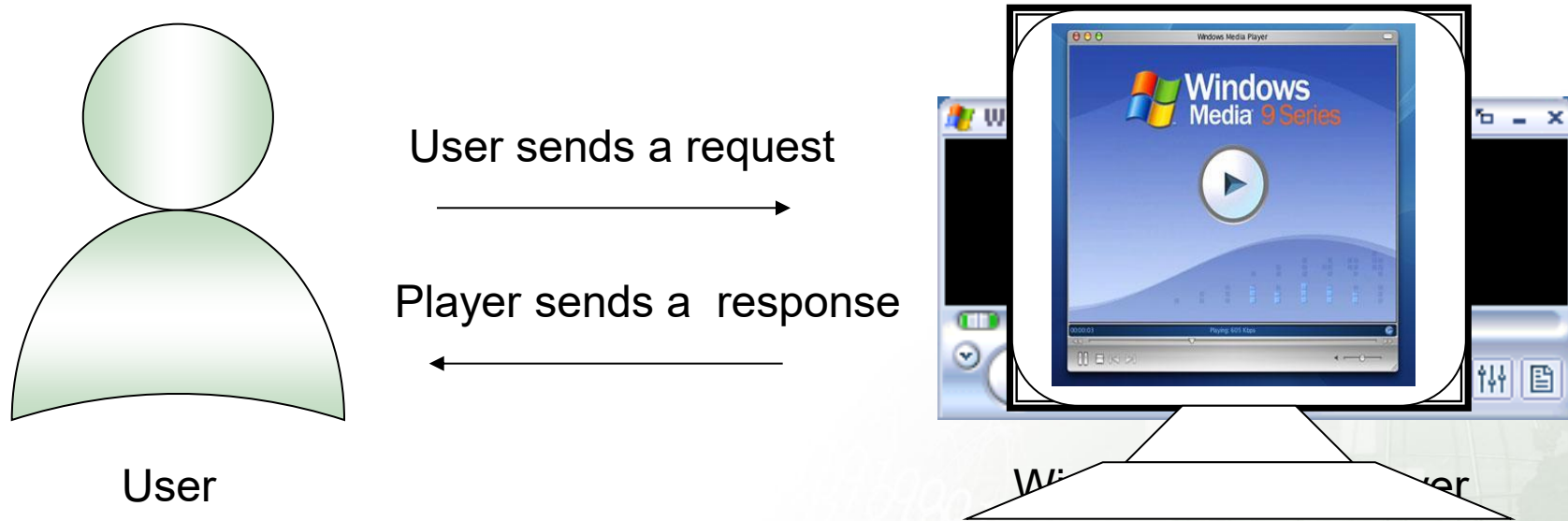


User



User selects an mp3 file

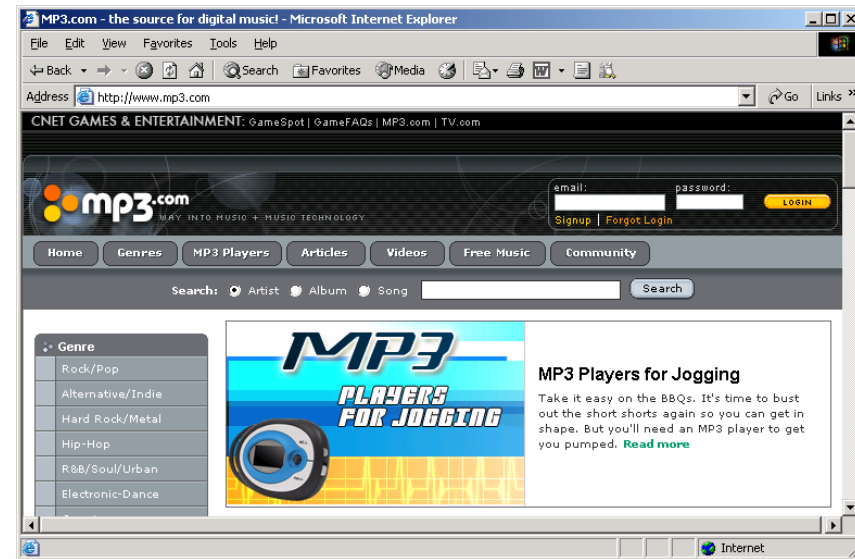
Desktop Applications 2-2



Media Player is an example of desktop application

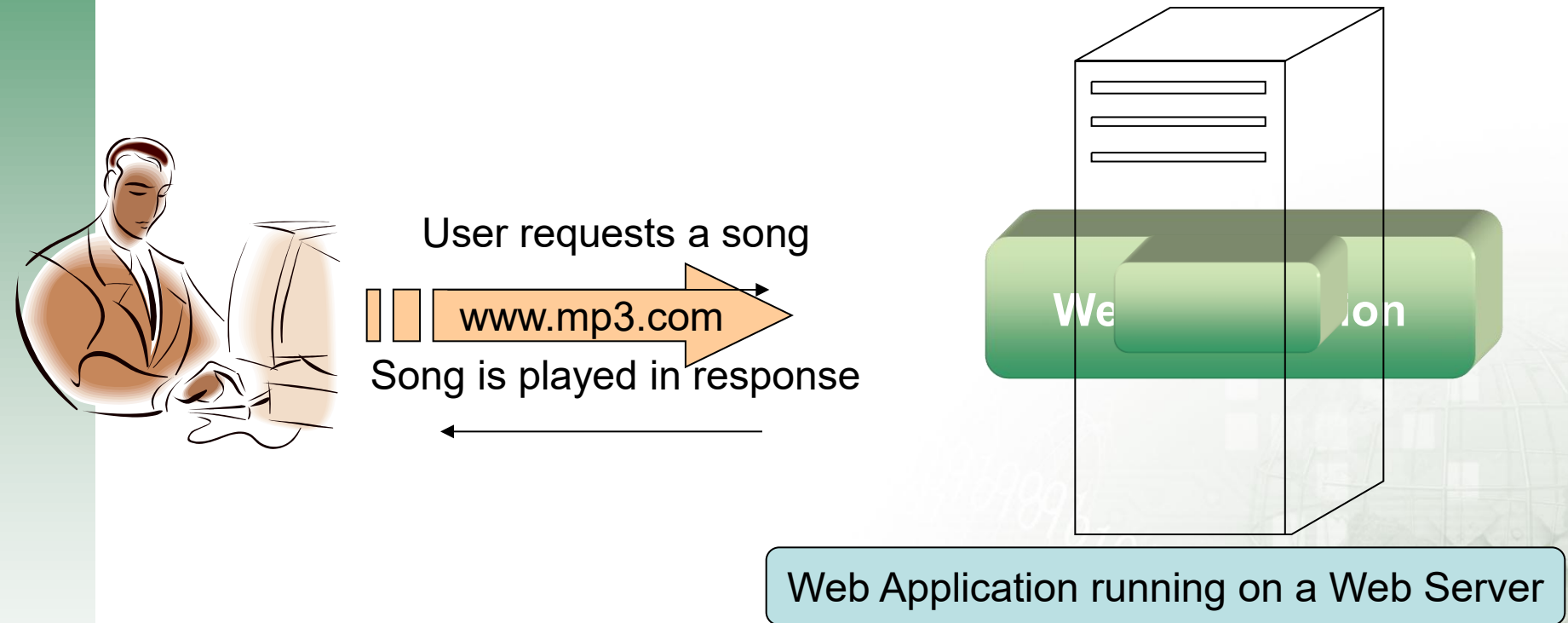
Response = Media Player plays the selected file

Web Applications 2-1



Web Site hosted on a
Web Server

Web Applications 2-2



Advantages of Web Applications:

- Accessing Web Applications is easier
- Maintenance and deployment costs are low

Introduction to Web

- ❑ Web is basically a system of *Internet* servers that supports formatted documents.
- ❑ The documents are formatted using a markup language called HTML (*HyperText Markup Language*) that supports links to other documents, like graphics, audio, and video files.



Web Intro

- ❑ Web consists of billions of clients and servers connected through wires and wireless networks.
- ❑ First, web clients make requests to a web server. Then, the web server receives the request, finds the resources and returns the response to the client.
- ❑ When a server answers a request, it usually sends some type of content to the client.
- ❑ Then, the client uses a web browser to send a request to the server.
- ❑ The server often sends a response back to the browser with a set of instructions written in HTML.
- ❑ All browsers know how to display HTML pages to the client.

Web & HTTP

- ❑ A website is a collection of static files i.e. web pages such as HTML pages, images, graphics etc.
- ❑ A Web application is a website with dynamic functionality on the server.
- ❑ Google, Facebook, Twitter are examples of web applications.

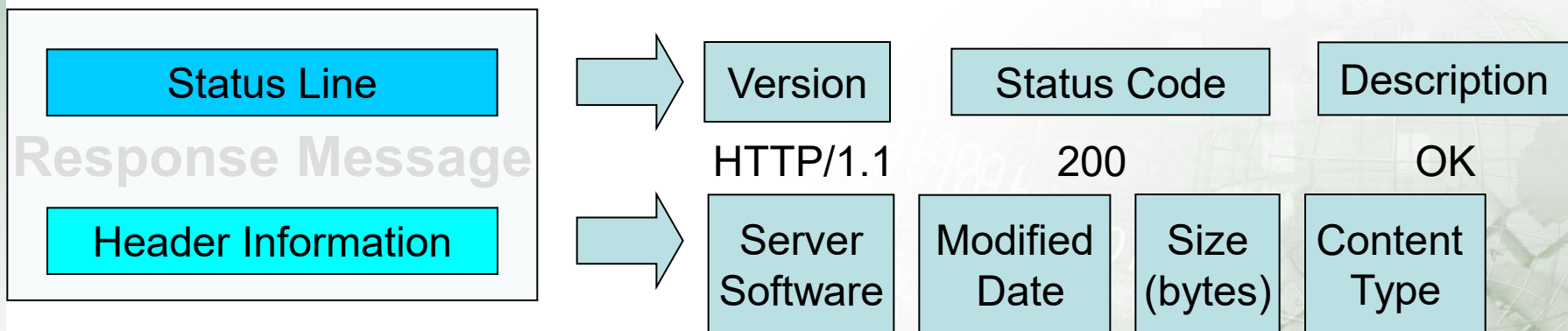
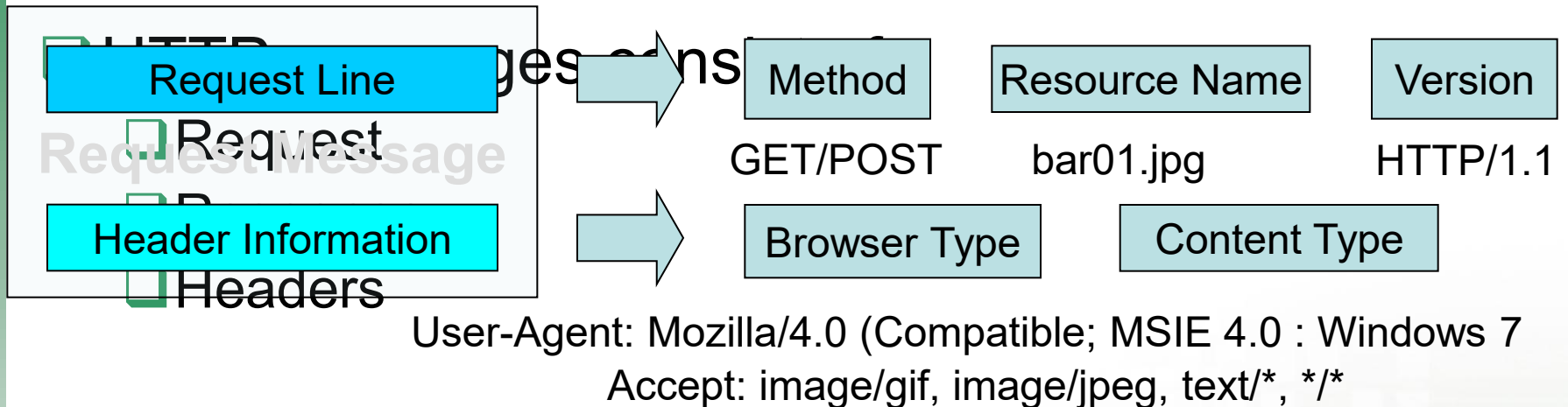


HTTP Protocol 3-1

- ❑ HyperText Transfer Protocol
 - ❑ Stateless protocol
 - ❑ Used to send request and response messages over the Internet.
 - ❑ Uses port 8080 to send and receive messages



HTTP Protocol 3-2

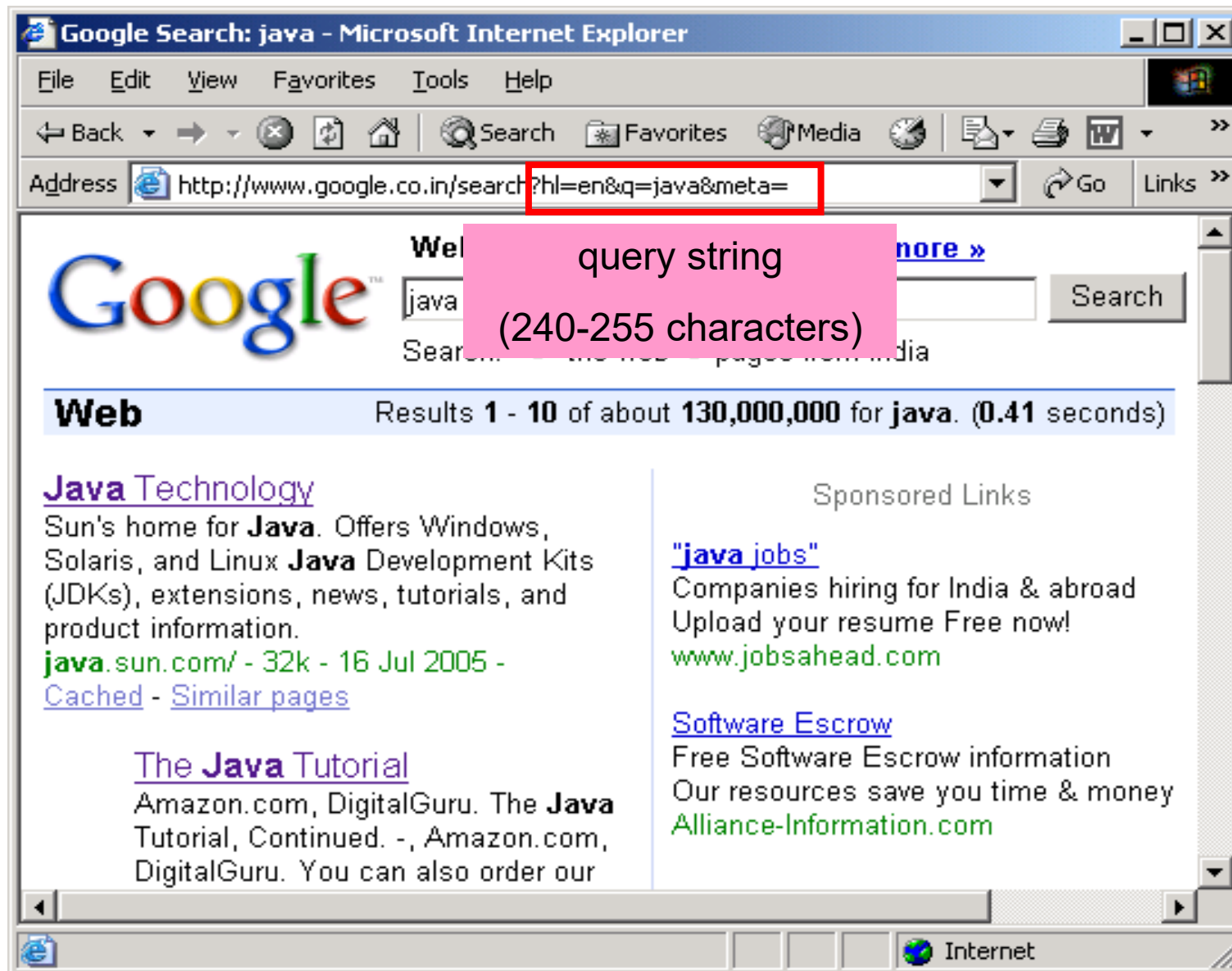


Server: Tomcat/ GlassFish / JBoss
Last modified: Tuesday, 07-Sep-21 1:14:34 GMT
Content-length: 100
Content-type: text/plain

HTTP Response Status Codes

Status Code	Constant	Meaning
200	SC_OK	Request successful
201	SC_CREATED	Resource created
302	SC_FOUND	Temporary redirect
400	SC_BAD_REQUEST	Invalid client request
401	SC_UNAUTHORIZED	Authentication required
403	SC_FORBIDDEN	Access denied
404	SC_NOT_FOUND	Resource not found
500	SC_INTERNAL_SERVER_ERROR	Server-side error

HTTP Protocol 3-3



HTTP

- ❑ HTTP is a protocol that clients and servers use on the web to communicate.
- ❑ It is similar to other internet protocols such as **SMTP** (Simple Mail Transfer Protocol) and **FTP**(File Transfer Protocol).
- ❑ HTTP is a *stateless protocol i.e* it supports only one request per connection.
- ❑ This means that with HTTP the clients connect to the server to send one request and then disconnect.
- ❑ This mechanism allows more users to connect to a given server over a period of time.
- ❑ The client sends an HTTP request and the server answers with an HTML page to the client, using HTTP.

Get and Post.

- ❑ The HTTP request can be made using a variety of methods, but the ones which we use widely are **Get** and **Post**.
- ❑ The method name itself tells the server the kind of request that is being made, and how the rest of the message will be formatted.

Get	Post
1. Data is sent in the header body 2. Restricted to limited data transfer 3. It is not secured 4. It can be bookmarked	1. Data is sent in the request body 2. Supports a large amount of data transfer 3. It is completely secured 4. It cannot be bookmarked

Application Architecture

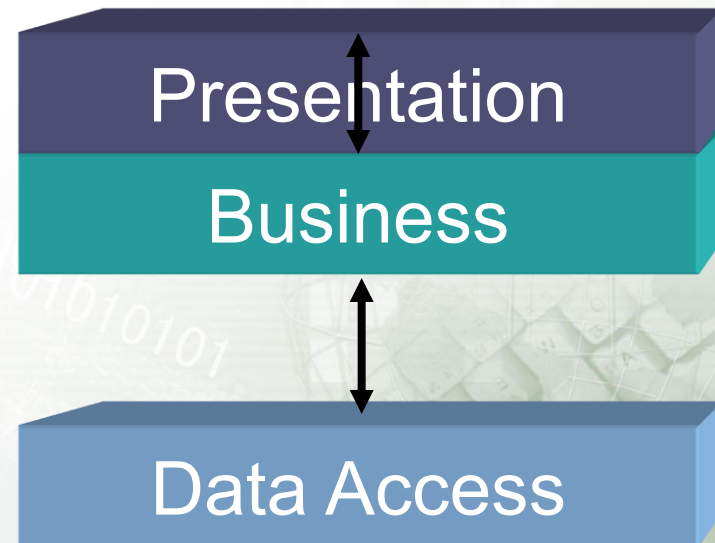
Software Application Components

- Presentation Logic
- Application Logic
- Data Access Logic



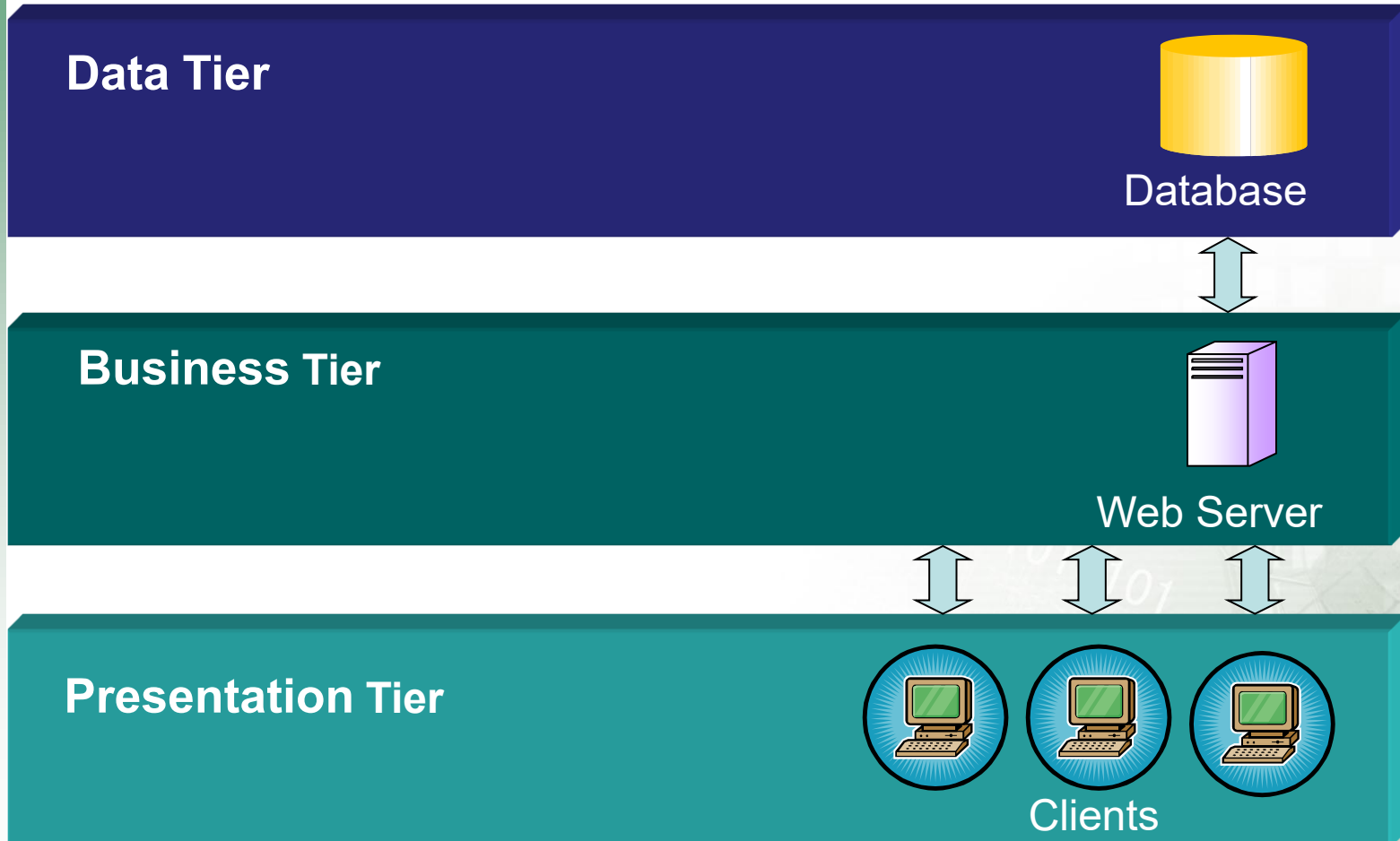
Types of Architecture

- One-Tier
- Two-Tier
- Three-Tier

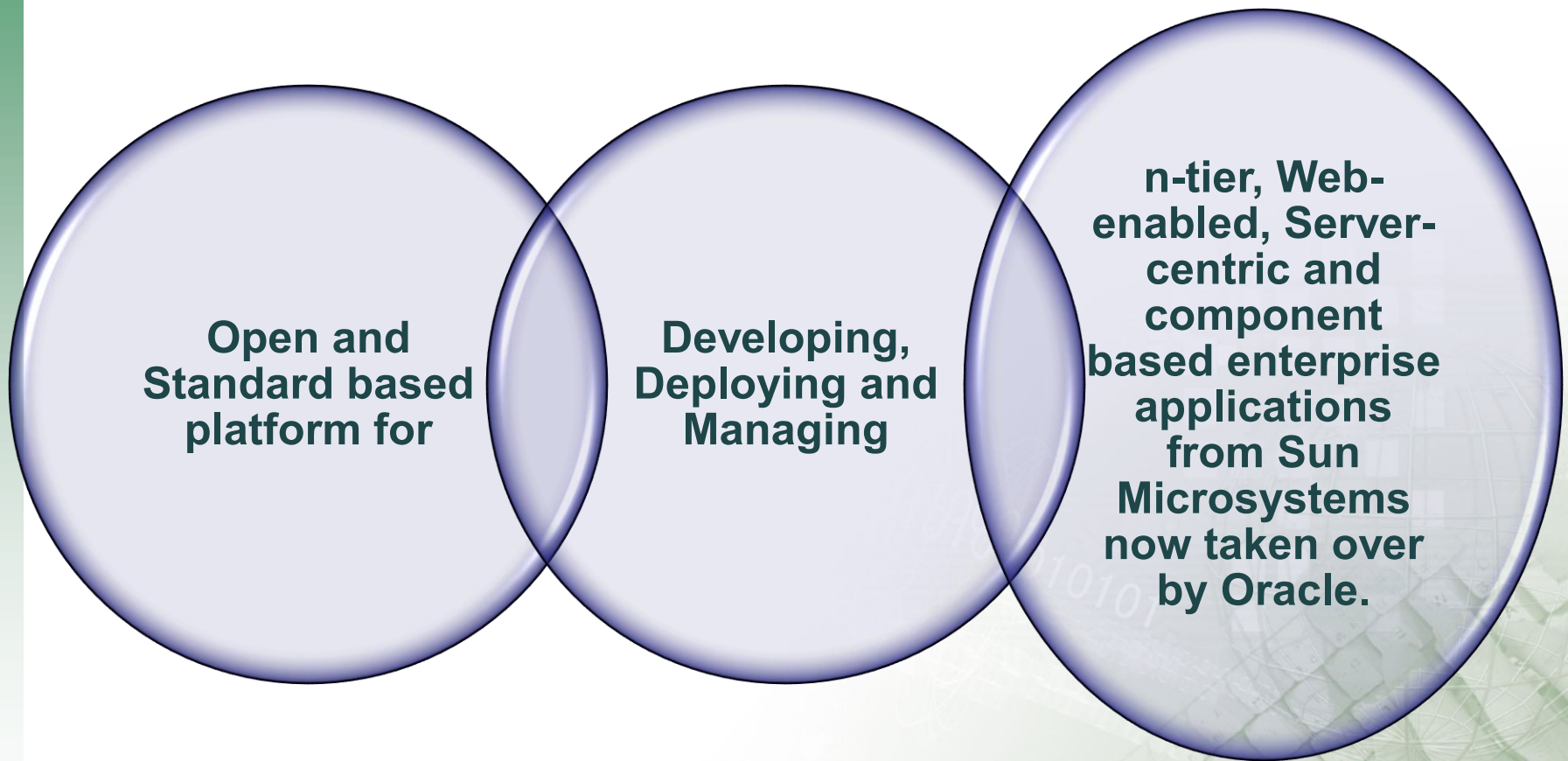


Web Application Architecture

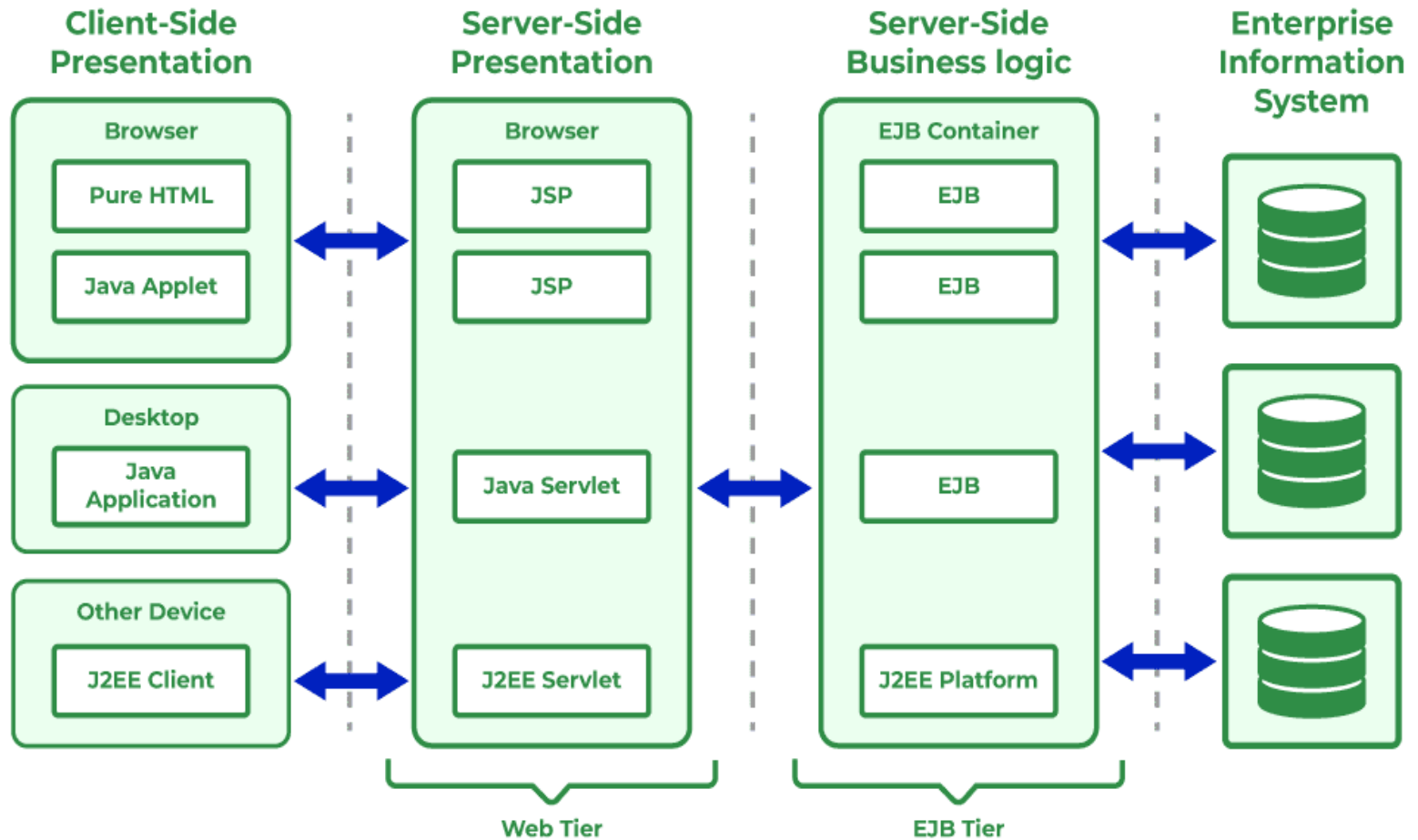
Web Applications use Three-tier architecture



What is J2EE ?



J2EE Components

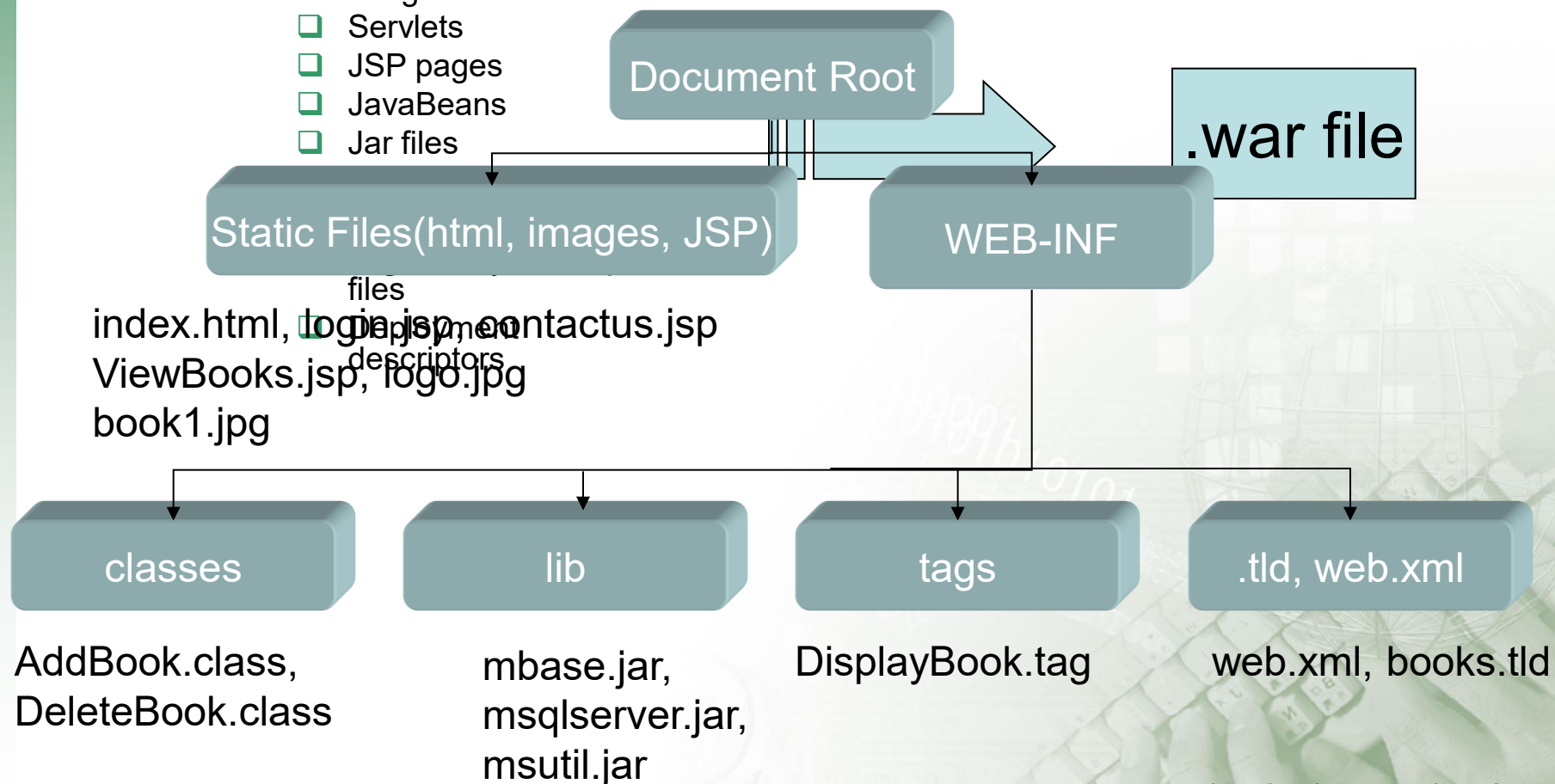


Web Application Directory Structure

Web Application

- HTML files
- Images
- Servlets
- JSP pages
- JavaBeans
- Jar files

Web Application Directory Structure



Web Application Development Process

- ☐ Design a directory structure
- ☐ Write web application code
- ☐ Write deployment descriptor
- ☐ Compile the code
- ☐ Package web application
- ☐ Deploy web application
- ☐ Execute web application

Server: Web vs. Application

- ❑ Server is a device or a computer program that accepts and responds to the request made by other program, known as client.
- ❑ It is used to manage the network resources and for running the program or software that provides services.

There are two types of servers:

- ❑ Web Server
- ❑ Application Server

Web Server

- ❑ Web server contains only web or servlet container.
- ❑ It can be used for servlet, jsp, struts, jsf etc.
- ❑ It can't be used for EJB.
- ❑ It is a computer where the web content can be stored.
- ❑ In general web server can be used to host the web sites but there also used some other web servers also such as FTP, email, storage, gaming etc.
- ❑ Examples of Web Servers are: Apache Tomcat and Resin.

A dynamic web server, in the context of web development, refers to a web server that can deliver content that changes based on user interactions, database inputs, or other real-time factors.

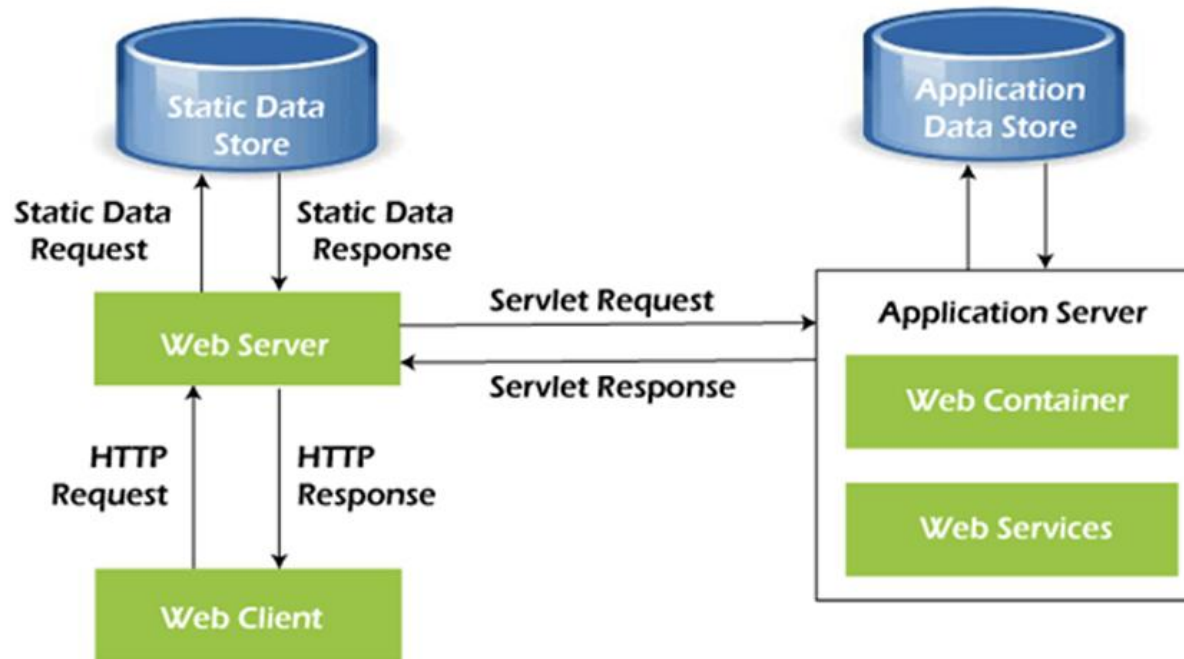
Web Server Working

It can respond to the client request in either of the following two possible ways:

- ❑ Generating response by using the script and communicating with database.
- ❑ Sending file to the client associated with the requested URL.

Block diagram representation of Web Server

Working of web servers



Application Server

- ❑ Application server contains Web and EJB containers.
- ❑ It can be used for servlet, jsp, struts, jsf, ejb etc.
- ❑ It is a component based product that lies in the middle-tier of a server centric architecture.
- ❑ It provides the middleware services for state maintenance and security, along with persistence and data access.
- ❑ It is a type of server designed to install, operate and host associated services and applications for the IT services, end users and organizations.
- ❑ Eg: Jboss, GlassFish, WebLogic, Web Sphere

An application server, also known as an app server or a web application server, is a type of server that hosts software used for business applications.

J2EE Containers

- ❑ Java web applications run in containers.
- ❑ A container is a program that manages the execution of web applications.
- ❑ It can be thought of as an operating system.
- ❑ OS manages the execution of any kind of application.
- ❑ The execution of Java web applications is managed by a container.

J2EE Containers

- ❑ Container program runs on the web server.
- ❑ So when a user requests a page, this message is forwarded to the container by the web server.
- ❑ Several steps carried out to process the user's request.
- ❑ Based on configuration settings, the container further invokes one of the servlets with objects to represent request message and response message.

J2EE Containers

- ❑ The container uses the information in request object and carries out all the operations required to process the request.
- ❑ After processing, the container sends the requested information back to the client using the response object.
- ❑ This is followed by releasing all the resources consumed by web application.

J2EE Containers

- ❑ The Application Server Which Controls And Provides Services Through An Interface Is Called A Container.
- ❑ The J2EE Container Helps In The Deployment And Execution Of Java Enterprise Edition(J2EE) Components.
- ❑ In J2EE There Are 5 Containers Types.
- ❑ Three Of These Are Server-Side Containers

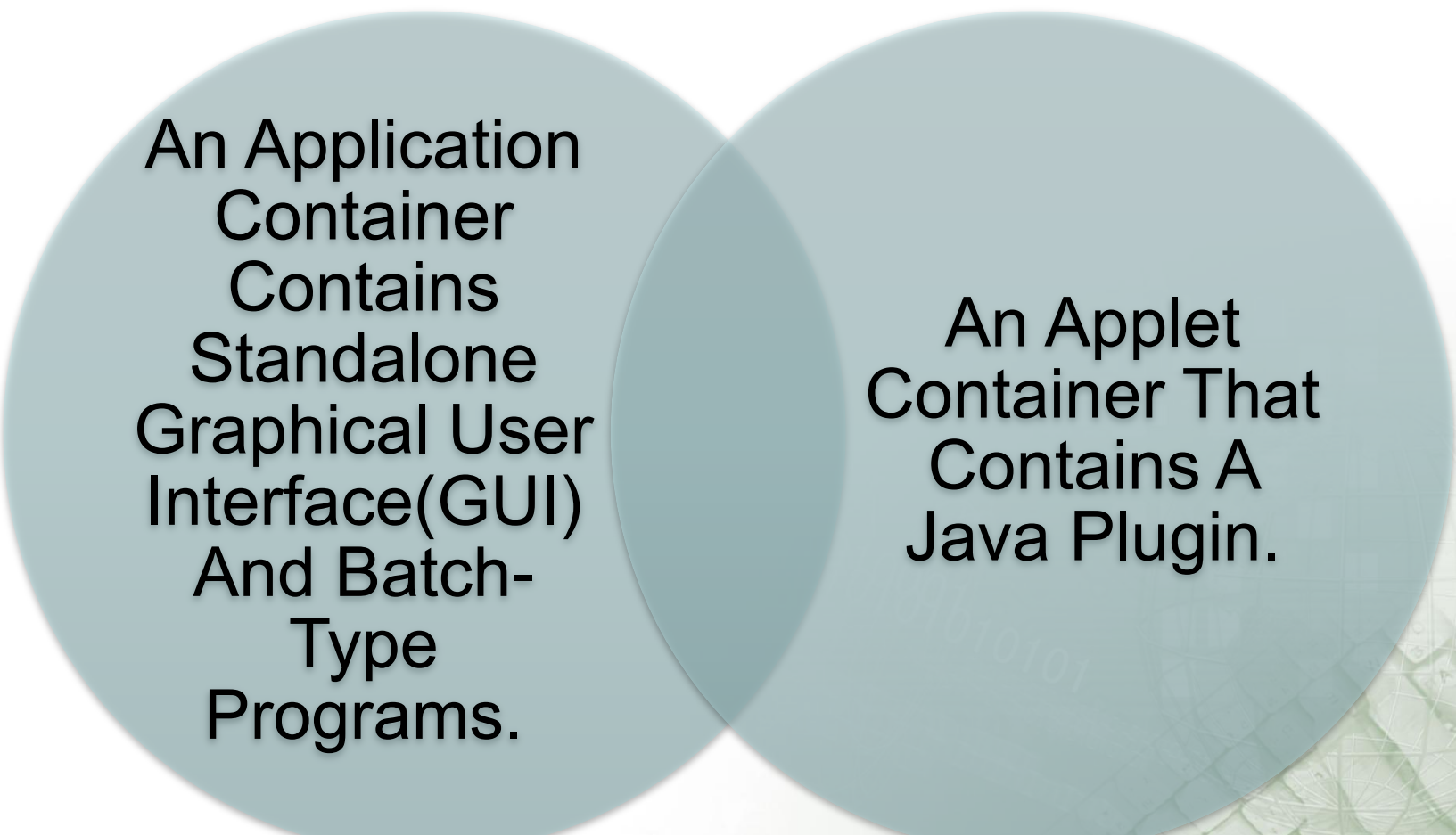
Server-Side Containers

The Server, Which Provides The Java Enterprise Edition(J2EE) Platform.

An EJB Container Contains (Enterprise Java Bean) EJB Components.

A Web Container That Manages Servlets.

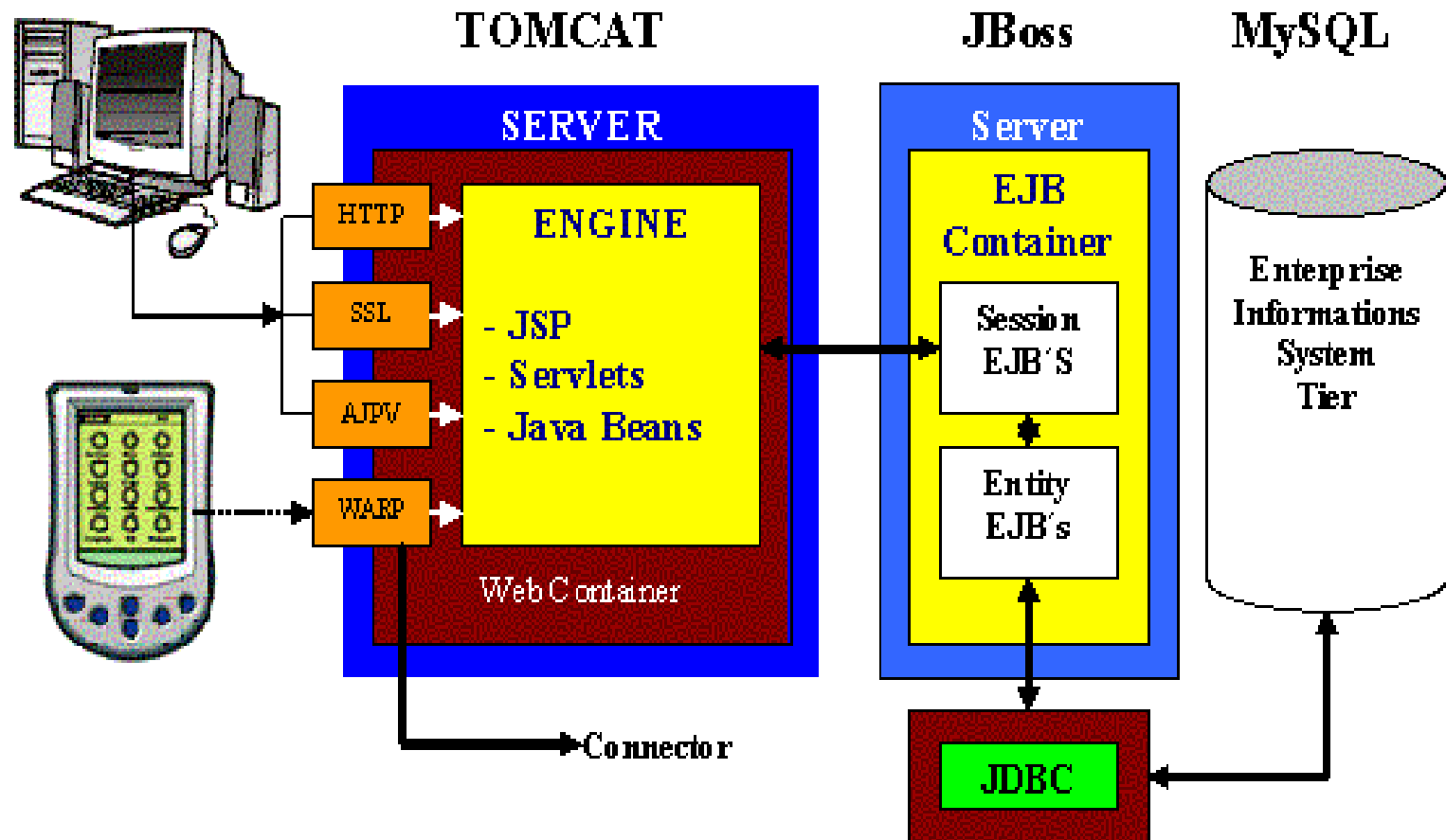
Two Containers on User-Side



An Application
Container
Contains
Standalone
Graphical User
Interface(GUI)
And Batch-
Type
Programs.

An Applet
Container That
Contains A
Java Plugin.

J2EE Container



J2EE Container Services

Invocation — Your Code Gets Loaded And Gets Run When Needed.

Transactional Context — Most Container Code Also Occurs In An Atomicity, Consistency, Isolation, Durability (ACID) Transaction Context.

Configuration — Things Like Java Database Connectivity(JDBC) Connections Are Thus Pass Up To You By The Container.

Security — The Container Will Also Manage Access To Your Code And Data To Authorized Users.

J2EE Container Services

Scalability — Since The Container Maintains It Can Schedule It And Automatically Fire Up Extra Copies If The Load Gets Heavy, Or, Can Be Statically Configured To Run Several Instances In Parallel.

Encapsulation — Your Program Exposes A Single Interface To The Container. However, Externally It May Expose This Interface In A Variety Of Forms.

Common Services — Such As Logging, Services Exposed By Other EJBs.

Tomcat Configuration

❑ Select Tools → Server Manager

Demonstration

Steps to configure Tomcat in Java Studio
Enterprise 8

Web.xml

- ❑ **Web.xml** defines mapping between URL paths and servlets that handle requests with those paths.
- ❑ The **web.xml** file provides configuration and deployment information for the Web components that comprise a Web application.
- ❑ The **web.xml** descriptor file represents the core of the java application.
- ❑ The web.xml file is located in **WEB-INF** directory of web application.

Example of web.xml File

```
<web-app>  
<servlet>  
  <servlet-name>HelloWorld</servlet-name>  
  <servlet-class>servletexample.createHelloWorldExample</servlet-  
  class>  
</servlet>  
<servlet-mapping>  
  <servlet-name>HelloWorld</servlet-name>  
  <url-pattern>/HelloworldExample/*</url-pattern>  
</servlet-mapping>  
</web-app>
```

Elements in Web.xml

- ❑ **<web-app>**: This element represents whole application of web.xml file
- ❑ **<servlet>**: This is the sub element of and represents the servlet
- ❑ **<servlet-name>**: This is the sub element of servlet and used to represents the name of servlet
- ❑ **<servlet-class>**: This is the sub element of servlet and used to represents the class of servlet
- ❑ **<servlet-mapping>**: This is the sub element of and used to map the servlet
- ❑ **<url-pattern>**: This is the sub element of servlet-mapping and used to client side to invoke the servlet

Web.xml Tags

- ❑ Welcome-file-list tag
- ❑ This tag is used to specify the default page of web application if none is specified.

Example:

```
<welcome-file-list>
```

```
<welcome-file>index.jsp</welcome-file>
```

```
<welcome-file>default.html</welcome-file>
```

```
</welcome-file-list>
```

- ❑ In the above example index.jsp used as web page for web application

Web.xml Tags

- ❑ Session config tag
- ❑ This tag is used to specify the session configuration parameter

Example:

```
<session-config>  
  <session-timeout>  
    15  
  </session-timeout>  
</session-config>
```

- ❑ In the above example session-config tag contains another tag session-timeout which specifies the http session timeout.
- ❑ The time specify in minutes.

Web.xml Tags

- ❑ Error page tag: This tag is used to specify the error occurred in the while weblogic server is responding ti a HTTP request, returns an HTML page that displays either the HTTP error code.

`<error-page>`

`<error-code>105</error-code>`

`<location>/jsp/error/PageNotFound.jsp</location>`

`</error page>`

- ❑ In the above example error-page tag specify the error code as 105 and location describes the location of the jsp page.

Advantages of web.xml file

- ❑ The first benefit of the xml is we can write it in our own markup language.
- ❑ There is no restriction to limited sets of tags.
- ❑ By defining our own tag we can create a markup language in terms of specific problem.
- ❑ Searching the data is easy and efficient.
- ❑ Completely compatible with Java™ and 100% portable.

Summary

- ❑ Web applications are easy to maintain, upgrade and deploy as compared to desktop applications.
- ❑ Pages requested using GET method can be bookmarked and emailed.
- ❑ POST method is used when sending sensitive information such as credit card numbers, passwords and so on.
- ❑ In three-tier architecture, the presentation tier cannot interact with the data tier.